

Challenges with Digitization from Aerial Photographs

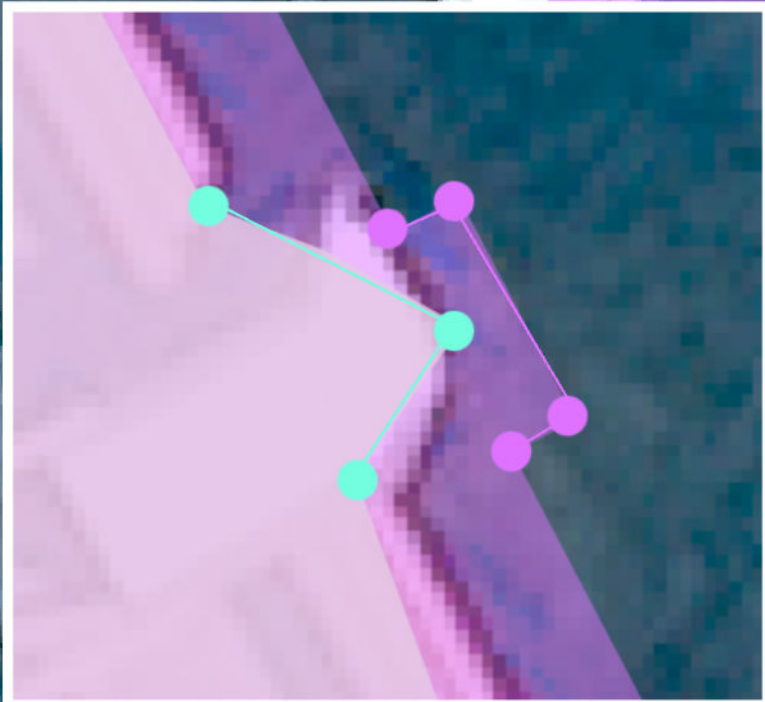
McGavin Building	Length (m)	Area (m2)	Vertices (num)
<i>Digitized Polygon</i>	160.31	1100.93	71
<i>Official Dataset</i>	160.06	1057.86	68
<i>Difference</i>	0.25	43.07	3

NUMBER AND POSITION OF VERTICES

Even slightly different position of vertices, or less vertices when manually digitalizing polygons can create drastically different geometries and thus dramatically different areas in the drawn polygon than from a ground-truthed dataset.

Polygon perimeter is largely unaffected by position of vertices but rather dramatically increases the more vertices are added.

Human error associated with manual digitization



● Vertices from Manual Drawing
● Vertices from Official Dataset

Building Footprint
(Visible from the Ground)

Top of the Building
(Visible from the Air)

RELIEF DISPLACEMENT

= a result of features of different heights in an image, the position of tall objects appears further away from the camera because of the angle of capture. The primary cause of positional inaccuracy.

■ Digitized Building Footprint
■ Official Building Footprint
■ Overlap Between Official and Digitized Footprints

0 0 0.01 0.02 KM